is equal and opposite to the change for the flea, so momentum conserved B1

4 **(a)** **(i)** angle subtended at centre of circle B1
(by) arc equal in length to the radius B1
(ii) arc = $r\theta$ and for one revolution, arc ≡ π (diameter) = $\pi(2r)$ M1
so, $\theta = \pi(2r) / r = 2\pi$ A0
(b) **(i)** point S shown vertically below C B1
(ii) [(max) force / tension – weight] provides the centripetal force C1
$18 - 3 = m\,r\,\omega^2 = (3 / 9.81)\,(0.85)\,\omega^2$ C1
$\omega = 7.6$ rad s^{-1} A1
(c) **(i)** vertically no net force, $T \cos 35^\circ = 3.0$, $T = 3.7$ N A1
(ii) resultant is horizontal component of tension
$3.7 \sin 35^\circ = 2.1$ N A1
horizontally towards the left B1

5 (a) (i) obeys the law pV/T = constant or any two named gas laws M1